

Key technology requirements for the Army, Navy and Air Force

Common technology requirements for the three forces

- Battlefield transparency
- Command and control architecture
- Communication systems
- Smart radios
- Information dominance
- Electronic warfare
- Unmanned systems
- Advance weapon systems
- EMP weapons
- Adaptive warheads
- Weapon guidance
- Space-based radar
- Stealth
- Digital systems
- Adaptive antenna signatures
- SAGW (Surface to Air Guided Weapon)
- Sensors
- Sensor fusion

Requirements for aviation

- Aircraft structures
- Propulsion technology
- Engine systems
- Avionics – displays, navigation capability, auto pilot systems, night/all weather capability, airborne sensors
- UAVs
- UCAVs
- Satellites, communications, meteorology
- Environmental friendly technology

Requirements for land warfare

- Future combat systems – mobility, lethality, survivability

- Guns, rockets, missile systems
- Robotics-based systems
- Biotechnology-based weapons
- Non-lethal weapons – Stun grenades, optimal weapons, acoustic weapons

Requirements for the Navy

- Ship launched/recoverable UAVs and UCAVs
- Laser-based wake detection capabilities
- LF transducers for under-water detection
- Multi-range super-cavitating torpedoes with active HH
- Development of ship-fitted guns and mountings, with limited surface-attack capability
- Towed array, variable-depth and sea-bed sonar systems
- Development of high-accuracy INS (Inertial Navigation System)
- Amphibious aircraft
- UAVs
- Airborne anti-submarine weapons

Marine engineering

- Main propulsion equipment, prime movers for power-generation equipment, auxiliary equipment, machinery-control systems
- Gas turbines, diesel engines
- Gear boxes
- Hull systems

Electrical engineering

- Gyros
- Auto-power-management systems
- Static-frequency converters
- Maintenance-free batteries

Submarines

- Equipment displays
- AA weapons

ject that may involve an expenditure of ₹ 300 billion. For the Navy, there is a very large component of strategic electronics on-board. At the coast guard level, the naval platforms would require both shore and platform-based radars, sonars, sensors, fire control, navigation and communication equipment. Large procurements are due for the MMRC (medium multi-role combat aircraft) with respect to the Air Force. Upgrades of the existing fleet are on the anvil, so a whole lot of new inductions into the Air Force are about to take place.

Aerospace and Defence (A&D) sector offers good sustainability

The A&D sector is a growth market, and those who enter it will surely be

around for 20 to 25 years. That's the kind of sustainability to be expected. This sector can be a new and profitable business for companies that are already into design, custom-built products, contract manufacturing or component manufacturing.

Manjunatha Hebbar, senior VP and head, strategic solutions, Infotech Enterprises Limited, informs that, "The aerospace and Defence market might be huge and very important, but getting into this industry is very tough as well. The entry barriers are extremely high. Yet, at the same time, sustainability is very high once an organisation successfully enters this sector."

Technology, all over the world, has been powered by demand from the strategic electronics domain. Dr Ajay Kumar, joint secretary, Department of

Electronics and IT, government of India, says, "The Defence sector requires patience, perseverance and determination to make it. But all those who have made it, unanimously believe that having attained the ability to participate in Defence tenders has been a good experience."

The recently released National Cyber Security policy, too, focuses on the indigenous development of cyber security products. Dr Vivek Lall, president and CEO, Reliance Industries Ltd, explains that some of the proposals in the Union budget for 2013-14 are very encouraging to attract large investments in domestic electronics manufacturing. These include a 15 per cent investment allowance, levying higher import duties on set-top boxes, zero customs duty on the import of semi-